

BOOK: HOW to PROVE it

6.1) 1) skipping Base cases

Inductive step: Assuming $0+1+2+\dots+k = \frac{k(k+1)}{2}$ — ①

Prove " " " $+k+1 = \frac{(k+1)(k+2)}{2}$ — ②

This is $\frac{k^2+3k+2}{2}$

From ①, adding $k+1$ gives on R.H.S:

$$\frac{k(k+1)}{2} + 2(k+1) = \frac{k^2+k+2k+2}{2} = \frac{k^2+3k+2}{2}$$

← go back.

6.1) 2) Assuming $1^2+2^2+\dots+k^2 = \frac{k(k+1)(2k+1)}{6}$ — ①

Prove $1^2+2^2+\dots+k^2+(k+1)^2 = \frac{(k+1)(k+2)(2k+3)}{6}$ — ②

$$\downarrow$$

$$\frac{2k^3+9k^2+13k+6}{6} \text{ — ③ } (\because \text{by multiplying})$$

Adding $(k+1)^2$ to ① on both sides. The R.H.S gives

$$\begin{aligned} & \frac{k(k+1)(2k+1)}{6} + (k^2+2k+1) \\ &= \frac{k(k+1)(2k+1) + 6k^2 + 12k + 6}{6} \\ &= \frac{2k^3+2k^2+k^2+6k+k+6k^2+12k+6}{6} \\ &= \frac{2k^3+9k^2+13k+6}{6} \end{aligned}$$

Now go back from ③ to reach goal.

6.1) 3) Assuming $0^3+1^3+2^3+\dots+k^3 = \left[\frac{k(k+1)}{2}\right]^2$ — ①

Prove " " " $+(k+1)^3 = \left[\frac{(k+1)(k+2)}{2}\right]^2$ — ②

The R.H.S of ② simplified is $k^4+6k^3+13k^2+12k+4$ — ③

Adding $(k+1)^3$ to ① gives on R.H.S

$$\begin{aligned} & \left[\frac{k(k+1)}{2}\right]^2 + (k+1)^3 \\ &= \frac{k^2(k^2+2k+1)}{4} + (k^3+3k^2+3k+1) \\ &= \frac{k^4+2k^3+k^2+4(k^3+3k^2+3k+1)}{4} \end{aligned}$$

— This is equal to ③
go back

$$\begin{aligned}
 6.1) 4) \quad n=1 &\rightarrow 1 \\
 n=2 &\rightarrow 1+3=4 = 2^2 \\
 n=3 &\rightarrow 1+3+5=9 = 3^2 \\
 n=4 &\rightarrow 1+3+5+7=16 = 4^2
 \end{aligned}$$

so pattern is n^2

Assuming $1+3+5+\dots+2k-1 = k^2$ — (1)

Adding $2k+1$ (next odd number)

$$1+3+5+\dots+2k-1+2k+1 = k^2+2k+1 = (k+1)^2$$

6.1) 6) By making pattern & looking at Question 1 & 5, the formula is $\frac{n(n+1)(n+2)(n+3)}{4}$

Assuming $0 \cdot 1 \cdot 2 + 1 \cdot 2 \cdot 3 + \dots + k(k+1)(k+2)$ — (1)

Prove $0 \cdot 1 \cdot 2 + 1 \cdot 2 \cdot 3 + \dots + k(k+1)(k+2) + \frac{(k+1)(k+2)(k+3)(k+4)}{4}$ — (2)

② becomes $(k^3+6k^2+11k+6)(k+4)$

$$\begin{aligned}
 &= \frac{k^4+6k^3+11k^2+6k+4k^3+24k^2+44k+24}{4} \\
 &= \frac{k^4+10k^3+35k^2+50k+24}{4} \quad \text{--- (3)}
 \end{aligned}$$

Adding $(k+1)(k+2)(k+3)$ to ①'s R.H.S gives

$$\frac{(k^2+k)(k+2)(k+3)}{4} + (k+1)(k+2)(k+3)$$

= ③ so go back from ③

6.1) 8) Assuming $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{1}{2k-1} - \frac{1}{2k} = \frac{1}{k+1} + \frac{1}{k+2} + \dots + \frac{1}{2k}$ — (1)

Prove $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{1}{2k-1} - \frac{1}{2k} + \frac{1}{2k+1} - \frac{1}{2k+2} = \frac{1}{k+2} + \frac{1}{k+3} + \dots + \frac{1}{2k+2}$ — (2)

Adding $\frac{1}{2k+1} - \frac{1}{2k+2}$ to ① gives

$$\begin{aligned}
 \text{L.H.S} &= \text{L.H.S of ①} = \frac{1}{k+1} + \underbrace{\frac{1}{k+2} + \frac{1}{k+3} + \dots + \frac{1}{2k} + \frac{1}{2k+1} - \frac{1}{2k+2}}_{\text{same terms as in ②}} \quad \text{--- (3)}
 \end{aligned}$$

We just add $\frac{1}{k+1} - \frac{1}{2k+2}$ which gives $+\frac{1}{2k+2}$

so ③ becomes $\frac{1}{k+2} + \frac{1}{k+3} + \dots + \frac{1}{2k} + \frac{1}{2k+1} + \frac{1}{2k+2}$. Hence proved

$$6.1) 9) (a) \text{ supp } 2 \mid (k^2 + k) \quad \rightarrow \quad 2 \mid [(k+1)^2 + (k+1)]$$

$$\text{so } \exists m \in \mathbb{Z} (2m = k^2 + k) \quad \text{--- (1)}$$

$$\exists l \in \mathbb{Z} [2l = (k+1)^2 + (k+1)] \quad \text{--- (2)}$$

From (2), we find $2l = (k+1)^2 + k+1$

$$2l = k^2 + 2k + 1 + k + 1$$

$$2l = (k^2 + k) + (2k + 2)$$

$$2l = 2m + 2(k+1)$$

Clearly $2 \mid k^2 + k$ & $2 \mid 2(k+1)$ so from here, let $l = \frac{2m + 2(k+1)}{2}$ to complete the proof.

$$(b) \text{ supp } 6 \mid (k^3 - k) \quad \rightarrow \quad 6 \mid [(k+1)^3 - (k+1)]$$

$$\exists m \in \mathbb{Z} (6m = k^3 - k)$$

$$\exists l \in \mathbb{Z} [6l = (k+1)^3 - (k+1)]$$

$$\text{Note that } (k+1)^3 - k - 1 = k^3 + 3k^2 + 3k + 1 - k - 1$$

$$= k^3 + 3k^2 + 2k$$

$$= k^3 - k + 3k^2 + 3k \quad (\because \text{doubling \& sub } k)$$

$$= (k^3 - k) + 3(k^2 + k)$$

$$= 6m + 6 \frac{(k^2 + k)}{2} = 6 \left[m + \frac{k^2 + k}{2} \right]$$

so let $l = m + \frac{k^2 + k}{2}$. By previous part (a), $2 \mid k^2 + k$ so $\frac{k^2 + k}{2} \in \mathbb{Z}$.

$$6.1) 19) a \in \mathbb{R}, a < 0$$

$$\forall n \in \mathbb{N} \left[(n \text{ is even} \rightarrow a^n > 0) \wedge (n \text{ is odd} \rightarrow a^n < 0) \right] \quad \text{--- (1)}$$

Base case: 0 is even & $a^0 = 1 > 0$.

Inductive step: supp (1) holds for $k \in \mathbb{N}$ so if k is even then $a^k > 0$

& if k is odd then $a^k < 0$. --- (2)

We have to prove $(k+1 \text{ is even} \rightarrow a^{k+1} > 0) \wedge (k+1 \text{ is odd} \rightarrow a^{k+1} < 0)$ --- (3)

supp $k+1$ is even, then k is odd. By (2) $a^k < 0$. Multiplying both sides by a & reversing direction (as $a < 0$), we get $a^k \cdot a > 0$

so $a^{k+1} > 0$.

Now supp $k+1$ is odd, then k is even, so by (2) $a^k > 0$. Multiplying

by a , $a^k \cdot a < 0$ (as $a < 0$ so reversing direction of inequality).

so $a^{k+1} < 0$.